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**Operating Instructions
for
PMT Tube
P1**

FEATURES

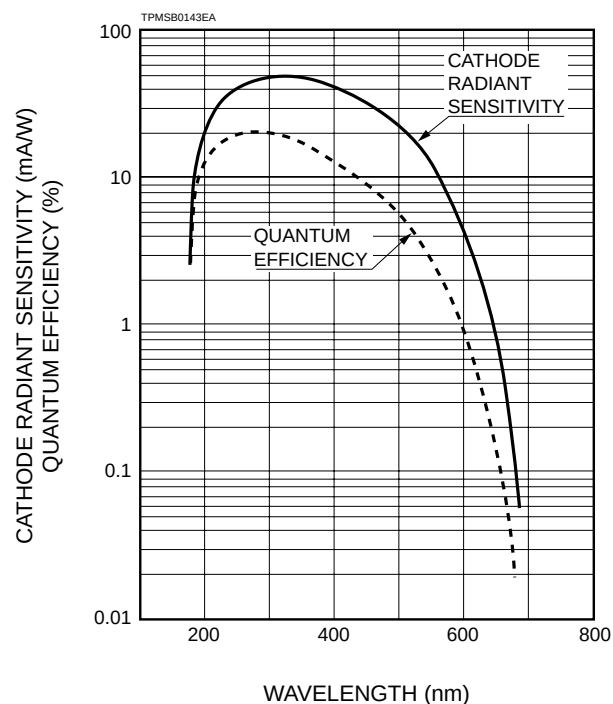
- 28mm (1-1/8 Inch) Diameter Side-on
- 185 to 650nm Spectral Response
- For UV to Visible Range Photocathode



GENERAL

Parameter		Description/Value	Unit
Spectral Response		185 to 650	nm
Wavelength of Maximum Response		340	nm
Photo-cathode	Material	Sb-Cs	—
	Minimum Useful Size	8 × 24	mm
Window Material		UV glass	—
Dynode	Structure	Circular-cage	—
	Number of Stages	9	—
Direct Interelectrode Capacitances	Anode to Last Dynode	4	pF
	Anode to All Other Electrodes	6	pF
Base		JEDEC No. B11-88	—
Suitable Socket		E678-11A	—

Figure 1: Typical Spectral Response



PHOTOMULTIPLIER TUBE 1P28

MAXIMUM RATINGS (Absolute Maximum Values)

Parameter		Value	Unit
Supply Voltage	Between Anode and Cathode	1250	Vdc
	Between Anode and Last Dynode	250	Vdc
Average Anode Current		0.1	mA
Ambient Temperature		-80 to +50	°C

CHARACTERISTICS (at 25°C)

Parameter		Min.	Typ.	Max.	Unit
Anode Sensitivity	Luminous (2856K)	20	400	—	A/lm
	Radiant at 400nm	—	4.8×10^5	—	A/W
Cathode Sensitivity	Luminous (2856K)	25	40	—	μA/lm
	Radiant at 400nm	—	48	—	mA/W
	Quantum Efficiency at 340nm	—	21	—	%
Gain		—	1.0×10^7	—	—
Anode Dark Current (after 30 minute)		—	5	50	nA
Time Response	Anode Pulse Rise Time	—	2.2	—	ns
	Electron Transit Time	—	22	—	ns

VOLTAGE DISTRIBUTION RATIO AND SUPPLY VOLTAGE

Electrodes	K	Dy1	Dy2	Dy3	Dy4	Dy5	Dy6	Dy7	Dy8	Dy9	P
Ratio	1	1	1	1	1	1	1	1	1	1	1

Supply Voltage: 1000Vdc, K: Cathode, Dy: Dynode, P: Anode

Figure 2: Typical Gain Characteristics

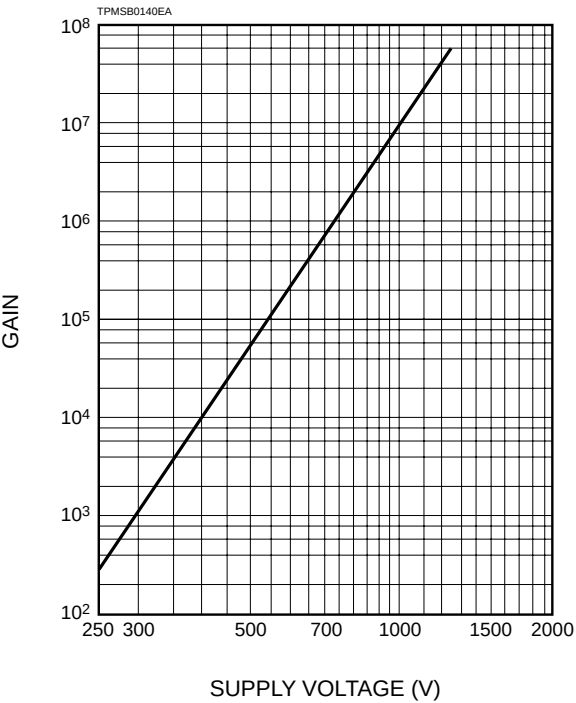
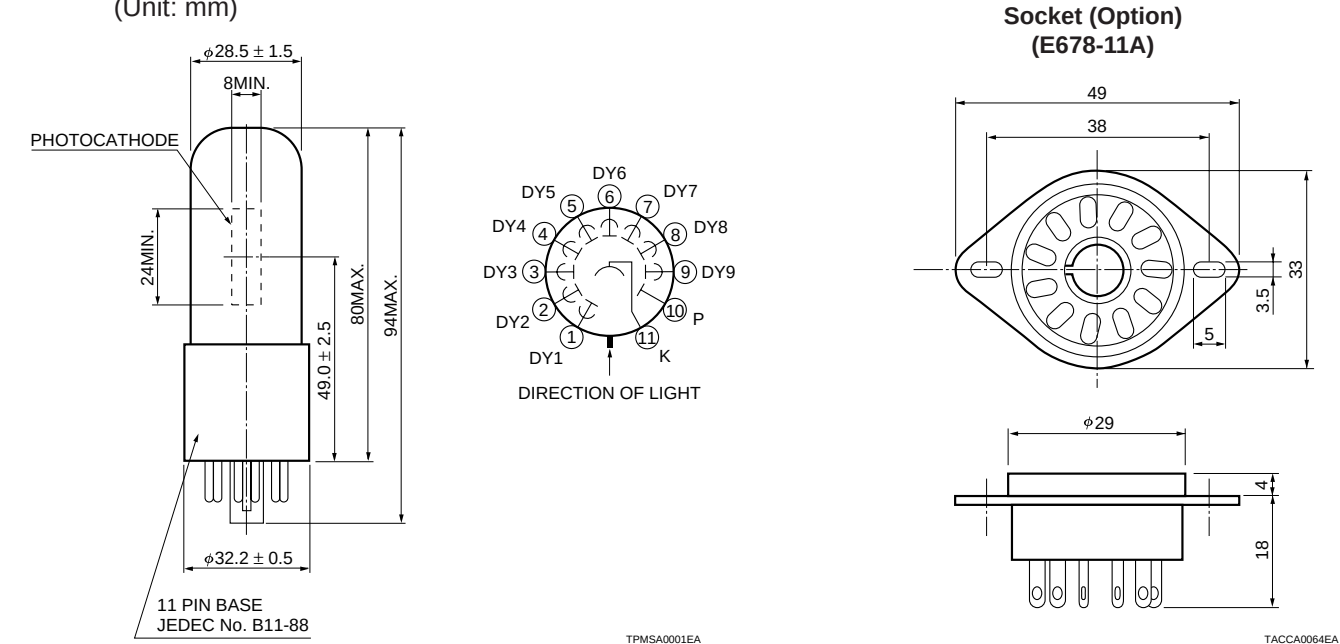


Figure 3: Dimensional Outline and Basing Diagram (Unit: mm)



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